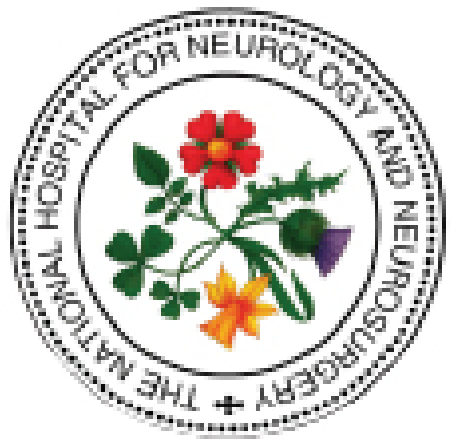




The National Hospital for Neurology and Neurosurgery Neuroscience Critical Care Unit (NCCU)

KETAMINE – POST-OPERATIVE LOW DOSE INTRAVENOUS ADMINISTRATION

May only be prescribed on advice of Anaesthetist or Acute Pain Team

(See NHNN Acute Pain Management Guidelines, Appendix 11)

Low dose ketamine infusion may be administered in Recovery, Neurocritical Care and wards at NHNN

INDICATIONS FOR USE

- Expected post-operative pain to be severe (following major complex spinal surgery)
- Expected postoperative pain to be moderate or major but patient is opioid tolerant or dependent (on high doses >100MME pre-operatively)
- Patients with acute exacerbation of chronic pain who present for moderate or major surgery
- Patients at high risk of opioid-related side effects (respiratory depression, obstructive sleep apnoea or severe post-operative nausea and vomiting), requiring opioid sparing techniques
- Patients at high risk of developing chronic post-operative pain (uncontrolled pre or postoperative pain, nerve damage, recent chemo or radiotherapy)

Do **NOT** give to patients in the following categories

- Following intracranial surgery
- Raised intracranial and intraocular pressure
- Poorly controlled cardiovascular disease including uncontrolled hypertension
- Severe liver disease (cirrhosis)
- Active psychosis
- History of ketamine abuse
- Pregnancy

ADVERSE EFFECTS

Adverse effects following anaesthetic doses of ketamine are listed in the BNF - *Anaesthetic doses are up to 60 x the amount given as low-dose intravenous infusion – therefore at 3mg/h likelihood of these side effects exceptionally rare*

- Common or very common:** Diplopia; hallucinations; hypertension; nausea; nightmares; nystagmus; rash; tachycardia; transient psychotic effects; vomiting
- Uncommon:** Arrhythmias; bradycardia; hypotension; laryngospasm; respiratory depression
- Rare:** Apnoea; cystitis; haemorrhagic cystitis; hypersalivation; insomnia
- Frequency not known:** Raised intra-ocular pressure

For any episode of adverse effects discontinue the ketamine infusion and contact on-call Anaesthetist (bleep 8131)to review patient - Datix incident report should be completed



**ANAESTHETIST or RECOVERY, NCC, ODP/ODA or ACUTE PAIN CNS)
RESPONSIBLE FOR PREPARING KETAMINE INFUSION**

STEP 1

Prepare 100mg IV ketamine made up to 50ml with 0.9% sodium chloride (2mg/ml) to run at **1.5mL/h** (3mg/h)

STEP 2

LOADING DOSE

Immediately prior to commencing infusion give a **3mg IV bolus** of ketamine

STEP 3

INFUSION

Commence IV infusion at **3mg/h (1.5ml/h)**
Continue ketamine infusion until syringe runs out or any medical reasons to stop infusion / or as prescribed up to 72 hours maximum

- Use giving-set with anti-syphon valve to administer ketamine infusion to prevent drug from free-flowing into patient
- To prevent back-tracking, anti-reflux valve, sometimes called non-return valve or back-check valve, should be used on extension lines where gravity set may be connected
- Patients with ketamine infusion may go to a neurosurgical ward as part of their multi-modal analgesia regime
- Syringe pump must be contained within a lockbox to prevent unauthorised access to the infusion
- May be administered concurrently with PCA in a PCA approved clinical area
- May be administered concurrently with morphine infusion in areas that administer morphine infusions

DISCONTINUATION OF INFUSION

- Discontinue after 24 hours or sooner if pain completely resolves or patient does not tolerate treatment
- If Anaesthetist or Acute Pain Team recommends continuation of low dose ketamine then a new syringe must be prepared every 24 hours
- It is not expected that the Ketamine infusion will run for more than 72 hours
- Ketamine infusion does not need to be weaned - Stop infusion and disconnect when prescription discontinued

